File 30: UGA Intern Workstream — Page 3: Riparian Buffer Vegetative Survey & Canopy Planning

> ***Ecology & Biological Monitoring Training Manual Target School****: University of Georgia (UGA) Odum School of Ecology / Warnell School of Forestry* ***Project Area****: Upper Savannah Basin & Chattahoochee River Spawning Headwaters* ***Supervising Board****: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Scientific Objective

In stream mitigation banking, physical bank stability and biological temperature control depend entirely on the establishing riparian forest buffer.
Under the **USACE Savannah District Standard Operating Procedure (SOP)**, a stream mitigation project must maintain a minimum **100-foot native vegetative buffer** recorded on county deeds. To verify that our plantings successfully establish and thrive, we must conduct structured vegetative surveys over a **7-year monitoring window**. Your primary role as a UGA Forestry or Ecology Intern is to layout monitoring plots, audit native plant survival rates, calculate tree densities, and manage invasive species eradication.

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2. Vegetative Monitoring Plot Design

You will deploy standardized **$10\text{m} \times 10\text{m}$ square vegetation plots** to collect statistically sound samples across our showcase tracts.

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                 10m x 10m VEGETATIVE MONITORING PLOT

     (0,10) __________________________________________ (10,10)
           |    * Tsuga canadensis (Hemlock)          |
           |                                          |
           |                     * Alnus serrulata    |
           |                       (Tag Alder)        |
           |   * Rhododendron maximum                 |
           |     (Geomorphic Root Bank)               |
           |                                          |
           |              * Cornus florida            |
           |                (Flowering Dogwood)       |
      (0,0) __________________________________________ (10,0)
                         [ Transect Center ]
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A. Layout Procedure

1. **Frequency**: Establish exactly **one plot per 300 linear feet** of restored stream bank corridor.
2. **Setup**: Using a fiberglass tape measure and a hand compass, measure a $10\text{m} \times 10\text{m}$ quadrat along the riparian zone. Mark the four corners with permanent orange PVC conduit pipe driven into

the ground, labeled with the plot identifier (e.g., VP-01, VP-02).

3. **Data Grid**: GPS-map the center of the plot using a sub-meter GIS receiver and record the coordinate in our database.

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3. Native Species Catalog & Target Densities

Within each monitoring plot, you will audit every planted tree and shrub. You must identify, count, and measure the height and stem diameter of the following native riparian species:

A. Core Riparian Woody Plantings

- **Rhododendron maximum (Great Laurel)**: Our primary high-gradient bioengineering tool. Deep fibrous root networks bind loose soils, while the evergreen leaves provide year-round shade.
- **Alnus serrulata (Tag Alder)**: High-growth woody shrub that stabilizes the stream bank toe-slopes. Actively fixes nitrogen to enrich poor soils.
- **Tsuga canadensis (Eastern Hemlock)**: Dominant coldwater canopy species. Requires active monitoring for Hemlock Woolly Adelgid (HWA) infestations.
- **Cornus florida (Flowering Dogwood)**: Excellent understory shrub that thrives on damp forest margins.
- **Betula nigra (River Birch)**: Fast-growing tree whose broad roots anchor unstable, high-sloped bank bends.

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4. Vegetative Survival & Stem Density Formulas

Under USACE criteria, our riparian buffers must satisfy two key milestones to release credit reserves:

1. Maintain a minimum tree density of **260 stems per acre** of native canopy.
2. Demonstrate an **$\ge 80\%$ survival rate** of initially planted woody species at Year 5.

You will calculate these values using the following equations on your field data sheets:

A. Stem Density Calculation

$$\text{Stems per Acre} = \frac{\text{Count of Live Native Stems in Plot}}{\text{Plot Area in Square Meters}} \times 4,047$$

Worked Example:
If you count exactly **8 live native woody plants** inside your $10\text{m} \times 10\text{m}$ *plot* (100 m^2):

$$\text{Stems per Acre} = \frac{8}{100} \times 4,047 = 323.76\text{ stems/acre}$$

Status: **Passes** the minimum 260 stems/acre threshold.

B. Percent Survival rate

$$\text{Percent Survival} = \frac{\text{Count of Live Planted Stems at Year } N}{\text{Count of Originally Planted Stems in Plot}} \times 100\%$$

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5. Invasive Species Management

The presence of non-native invasive species threatens native biodiversity and compromises buffer stability. You will conduct active eradication campaigns to keep total invasive coverage ***<5\%*** across the entire project area.

A. Primary Invasive Targets

- **Ligustrum sinense (Chinese Privet)**: Vigorous woody shrub that out-competes native tag alders and block hemlock saplings.
- **Hedera helix (English Ivy)**: Evergreen vine that climbs canopy trees, suffocating leaves and adding wind load that causes tree falls.
- **Microstegium vimineum (Japanese Stiltgrass)**: Aggressive annual grass that smothers the forest floor, preventing native seed germination.

B. Eradication Protocol

- **Physical**: Hand-pull young stiltgrass and privet before seed-set in late summer. Ensure the complete root systems are removed to prevent resprouting.
- **Chemical (Buffered)**: For mature Chinese Privet, apply a targeted **stem-cut treatment** utilizing a **50% water-soluble Glyphosate formulation** (approved for aquatic environments, such as Rodeo) applied directly to the cut trunk cambium within 5 minutes of cutting. Never spray broad herbicides near open flowing waters!